Fundamental Principles of Physics

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1 Motion

Physics is the study of motion and change. In mechanics, the motion of individual objects is studied. In other domains, such as electricity and magnetism, we focus more on fields and on flows of particles, charge, momentum, and energy. In essence, these domains too can be regarded as the study of change in space and time.

In mechanics, we model motion as a function that assigns a position in space to every instant of time:

$$\boldsymbol{r}(t) = x(t)\hat{\boldsymbol{\imath}} + y(t)\hat{\boldsymbol{\jmath}} + z(t)\hat{\boldsymbol{k}}.$$

Here, $\boldsymbol{r}(t)$ is the position vector from the origin of the chosen coordinate system to the position at time t. The set of these positions forms the spatial trajectory. Motion is therefore described by a scalar function of t for each spatial dimension.

We assume that $\boldsymbol{r}(t)$ is differentiable at least once. We call its first derivative the velocity vector:

$$\boldsymbol{r}\,'(t) = x'(t)\hat{\boldsymbol{\imath}} + y'(t)\hat{\boldsymbol{\jmath}} + z'(t)\hat{\boldsymbol{k}}.$$

The velocity vector has both a magnitude and a direction.

Motion along a fixed straight line has the parametrization

$$\boldsymbol{r}(t) = \boldsymbol{r}_0 + s(t)\hat{\boldsymbol{u}},$$

where $\boldsymbol{r}_0$ is a fixed point on the line and $\hat{\boldsymbol{u}}$ is a constant unit vector indicating its direction. The direction of the trajectory is constant, but the speed at which the trajectory is traversed may vary with $s(t)$. Indeed,

$$\boldsymbol{r}'(t) = s'(t)\hat{\boldsymbol{u}}$$

and

$$\boldsymbol{r}''(t) = s''(t)\hat{\boldsymbol{u}}.$$

A subtle misunderstanding can arise here. For one-dimensional motion, we often draw the graph $(t, x(t))$. When this graph is a straight line,

$$x(t) = x_0 + vt,$$

where v is constant. A straight line in the graph $(t, x(t))$ therefore means that the velocity is constant.

This must be distinguished from a straight spatial trajectory. A straight spatial trajectory

$$\boldsymbol{r}(t) = \boldsymbol{r}_0 + s(t)\hat{\boldsymbol{u}}$$

may be traversed with a variable speed.

When we consider time and space jointly in the worldline

$$\boldsymbol{R}(t) = \big(t, x(t), y(t), z(t)\big),$$

then, for a straight worldline,

$$\boldsymbol{R}(t) = \boldsymbol{R}_0 + t\boldsymbol{W},$$

where

$$\boldsymbol{W} = (1, v_x, v_y, v_z)$$

is constant. Only for a straight worldline are both the spatial direction and the spatial velocity constant.

One of the fundamental laws of physics is Galileo's law of inertia. It states that motion without external interaction manifests itself as motion along a straight line with constant velocity in space, and therefore as a straight worldline.

Galileo's law therefore states that free motion is determined by two initial conditions: position and velocity. If we write the trajectory as a Taylor polynomial, this becomes

$$\boldsymbol{r}(t) = \sum_{n=0}^{\infty} \boldsymbol{a}_n (t - t_0)^n,$$

with $\boldsymbol{a}_2 = \boldsymbol{a}_3 = \boldsymbol{a}_4 = \cdots = \boldsymbol{0}$.

The law states that a change in the magnitude or direction of the velocity ($\boldsymbol{a}_2 \neq \boldsymbol{0}$) requires an interaction.

For general motion, we write the velocity vector as

$$\boldsymbol{r}'(t) = v(t)\hat{\boldsymbol{T}}(t),$$

where $v(t)$ is the magnitude of the velocity and $\hat{\boldsymbol{T}}(t)$ is the unit tangent vector to the trajectory. Differentiation gives

$$\boldsymbol{r}''(t) = v'(t)\hat{\boldsymbol{T}}(t) + v(t)\hat{\boldsymbol{T}}'(t).$$

The acceleration may therefore consist of a change in the magnitude of the velocity, a change in direction, or both.

When $s(t)$ represents the arc length travelled,

$$v(t) = s'(t).$$

This gives

$$\boldsymbol{r}''(t) = s''(t)\hat{\boldsymbol{T}}(t) + s'(t)\hat{\boldsymbol{T}}'(t).$$

Because $\boldsymbol{r}''(t) = \boldsymbol{0}$ defines free motion according to Galileo, non-free motion must obey an equation of motion that determines $\boldsymbol{r}''(t)$:

$$\boldsymbol{r}''(t) = \boldsymbol{\psi}\left(t, \boldsymbol{r}(t), \boldsymbol{r}'(t)\right).$$

The function ψ describes how interactions locally change the motion. Together with the initial conditions

$$\boldsymbol{r}(t_0) = \boldsymbol{r}_0$$

and

$$\boldsymbol{r}'(t_0) = \boldsymbol{v}_0$$

the equation of motion determines the trajectory. The trajectory is thus the result of local dynamics.

When the equation of motion and its solution are locally analytic, the trajectory around t_0 can be written as a Taylor series:

$$\boldsymbol{r}(t) = \sum_{n=0}^{\infty} \boldsymbol{a}_n (t - t_0)^n.$$

The first two coefficients are determined by the initial conditions:

$$\boldsymbol{a}_0 = \boldsymbol{r}_0$$

and

$$\boldsymbol{a}_1 = \boldsymbol{v}_0.$$

Furthermore,

$$\boldsymbol{r}''(t) = \sum_{n=0}^{\infty} (n+2)(n+1)\boldsymbol{a}_{n+2}(t - t_0)^n.$$

Substituting the Taylor series into the equation of motion and equating the coefficients of equal powers of $t - t_0$ yields a recurrence relation for

$$\boldsymbol{a}_2, \boldsymbol{a}_3, \boldsymbol{a}_4, \ldots.$$

The initial conditions therefore determine the coefficients of orders zero and one. The equation of motion then recursively determines all coefficients from order two onward. The Taylor coefficients thus form the local trace of the trajectory, while the equation of motion is the rule that generates this trace.

2 Inertial Frames of Reference

Motion is always described relative to a chosen frame of reference. The position vector of an object therefore depends on the origin, the orientation of the axes, and the time measurement of that frame.

Consider two frames of reference S and S'. The origin of S', measured from S, moves according to

$$\boldsymbol{r}_{S'/S}(t) = \boldsymbol{r}_0 + \boldsymbol{u}t,$$

where $\boldsymbol{r}_0$ is the position of the origin of S' at time $t = 0$, and $\boldsymbol{u}$ is the constant velocity of S' relative to S.

We assume that the axes of both frames have the same orientation and do not rotate relative to one another. The position of an object measured from S' is then equal to its position measured from S, minus the position of the origin of S':

$$\boldsymbol{r}_{S'}(t) = \boldsymbol{r}_S(t) - \boldsymbol{r}_{S'/S}(t).$$

After substitution,

$$\boldsymbol{r}_{S'}(t) = \boldsymbol{r}_S(t) - \boldsymbol{r}_0 - \boldsymbol{u}t.$$

This is the Galilean transformation for position.

Differentiating with respect to time gives

$$\frac{d\boldsymbol{r}_{S'}}{dt} = \frac{d\boldsymbol{r}_S}{dt} - \boldsymbol{u}.$$

The velocity of an object therefore depends on the chosen frame of reference. Because $\boldsymbol{u}$ is constant, differentiating once more gives

$$\frac{d^2\boldsymbol{r}_{S'}}{dt^2} = \frac{d^2\boldsymbol{r}_S}{dt^2}.$$

The acceleration is therefore the same in all frames of reference that differ only by an initial position and a constant relative velocity.

Now consider an object on which no interaction acts. According to Galileo's law of inertia, in S

$$\frac{d^2\boldsymbol{r}_S}{dt^2} = \boldsymbol{0}.$$

Because the acceleration does not change in the transformation to S', we also have

$$\frac{d^2\boldsymbol{r}_{S'}}{dt^2} = \boldsymbol{0}.$$

The law of inertia therefore remains valid under the transformation from S to S'.

This can also be derived directly from the trajectory. Suppose that free motion in S is described by

$$\boldsymbol{r}_S(t) = \boldsymbol{r}_1 + \boldsymbol{v}t,$$

where $\boldsymbol{r}_1$ and $\boldsymbol{v}$ are constant. Then, in S',

$$\begin{aligned}\boldsymbol{r}_{S'}(t) &= \boldsymbol{r}_S(t) - \boldsymbol{r}_0 - \boldsymbol{u}t\\ &= \boldsymbol{r}_1 + \boldsymbol{v}t - \boldsymbol{r}_0 - \boldsymbol{u}t\\ &= (\boldsymbol{r}_1 - \boldsymbol{r}_0) + (\boldsymbol{v} - \boldsymbol{u})\,t.\end{aligned}$$

If we define

$$\boldsymbol{r}_{1,S'} = \boldsymbol{r}_1 - \boldsymbol{r}_0$$

and

$$\boldsymbol{v}_{S'} = \boldsymbol{v} - \boldsymbol{u},$$

then

$$\boldsymbol{r}_{S'}(t) = \boldsymbol{r}_{1,S'} + \boldsymbol{v}_{S'}t.$$

Thus, from S' as well, the object moves along a straight line with constant velocity.

The role of inertial frames of reference becomes even clearer when we write the trajectory as a Taylor series:

$$\boldsymbol{r}_S(t) = \boldsymbol{a}_0 + \boldsymbol{a}_1 t + \boldsymbol{a}_2 t^2 + \boldsymbol{a}_3 t^3 + \cdots .$$

Under the Galilean transformation

$$\boldsymbol{r}_{S'}(t) = \boldsymbol{r}_S(t) - \boldsymbol{r}_0 - \boldsymbol{u}t$$

only the coefficients of orders zero and one change:

$$\boldsymbol{a}_{0,S'} = \boldsymbol{a}_0 - \boldsymbol{r}_0$$

and

$$\boldsymbol{a}_{1,S'} = \boldsymbol{a}_1 - \boldsymbol{u}.$$

For all higher-order coefficients,

$$\boldsymbol{a}_{n,S'} = \boldsymbol{a}_n, \qquad n \geq 2.$$

A change of inertial frame therefore changes the described initial position and initial velocity, but not the coefficients that encode the change in motion.

Consider a general second-order equation of motion:

$$\frac{d^2\boldsymbol{r}_S}{dt^2} = \boldsymbol{\psi}\left(t, \boldsymbol{r}_S(t), \frac{d\boldsymbol{r}_S}{dt}\right).$$

The left-hand side remains unchanged under the Galilean transformation. If the right-hand side depends only on properties of the interaction that are likewise preserved under transformations between inertial frames, the equation of motion has the same form in both frames.

For two objects, for example, their relative position and relative velocity are invariant:

$$\boldsymbol{r}_{1,S'}(t) - \boldsymbol{r}_{2,S'}(t) = \boldsymbol{r}_{1,S}(t) - \boldsymbol{r}_{2,S}(t)$$

and

$$\frac{d\boldsymbol{r}_{1,S'}}{dt} - \frac{d\boldsymbol{r}_{2,S'}}{dt} = \frac{d\boldsymbol{r}_{1,S}}{dt} - \frac{d\boldsymbol{r}_{2,S}}{dt}.$$

Dynamics can therefore be described by the same equations of motion in all inertial frames of reference.

The equivalence of inertial frames of reference therefore need not be introduced as a separate principle. It follows from the fact that the transformation between such frames changes only the terms of orders zero and one, while the dynamical change from second order onward remains unchanged.

3 Mass as a Scale Factor of Motion

Up to this point, we have described motion without using the concept of mass. The trajectory

$$\boldsymbol{r}(t)$$

and its derivatives

$$\boldsymbol{v}(t) = \frac{d\boldsymbol{r}}{dt}$$

and

$$\boldsymbol{a}(t) = \frac{d^2\boldsymbol{r}}{dt^2}$$

describe the motion purely kinematically. They indicate where an object is located, how rapidly its position changes, and how that velocity changes. These quantities, however, do not yet distinguish between different amounts of matter undergoing the same motion.

Two objects may, for example, have the same velocity,

$$\boldsymbol{v}_1 = \boldsymbol{v}_2,$$

while their motion does not have the same physical scale. To specify that scale, we introduce the mass m.

We define mass as the scale factor by which the velocity of matter is converted into a quantity of motion. We call this quantity of motion momentum:

$$\boldsymbol{p} = m\boldsymbol{v}.$$

Velocity thus describes the motion itself, while mass indicates how much motion corresponds to that velocity. For two objects with the same velocity,

$$\boldsymbol{p}_1 = m_1\boldsymbol{v}$$

and

$$\boldsymbol{p}_2 = m_2\boldsymbol{v}.$$

When

$$m_1 > m_2,$$

it follows that

$$\|\boldsymbol{p}_1\| > \|\boldsymbol{p}_2\|.$$

Mass is therefore not required to define motion, but it is required to determine the physical scale of that motion. It connects the kinematic quantity velocity to the dynamical quantity momentum.

An interaction then manifests itself as a change in the quantity of motion. We write

$$\boldsymbol{F} = \frac{d\boldsymbol{p}}{dt}.$$

When the mass is constant,

$$\frac{d\boldsymbol{p}}{dt} = \frac{d}{dt}(m\boldsymbol{v}) = m\frac{d\boldsymbol{v}}{dt}.$$

Since

$$\frac{d\boldsymbol{v}}{dt} = \boldsymbol{a},$$

it follows that

$$\boldsymbol{F} = m\boldsymbol{a}.$$

The familiar inertial meaning of mass thus follows from its more fundamental role as a scale factor of the quantity of motion. For the same interaction,

$$\boldsymbol{a} = \frac{\boldsymbol{F}}{m}.$$

A greater mass therefore means that the same interaction causes a smaller change in velocity. Mass can thus be understood as the scale by which matter carries its motion and responds to an interaction.

This view connects directly to the Taylor expansion of the trajectory. Around an initial time t_0, we write

$$\boldsymbol{r}(t) = \boldsymbol{a}_0 + \boldsymbol{a}_1(t - t_0) + \boldsymbol{a}_2(t - t_0)^2 + \cdots .$$

Here,

$$\boldsymbol{a}_0 = \boldsymbol{r}(t_0)$$

and

$$\boldsymbol{a}_1 = \boldsymbol{v}(t_0).$$

The momentum at the initial time is then

$$\boldsymbol{p}(t_0) = m\boldsymbol{a}_1.$$

The interaction then determines the change in this scaled motion. For constant mass, at t_0,

$$\boldsymbol{F}(t_0) = m\boldsymbol{r}''(t_0).$$

Since

$$\boldsymbol{r}''(t_0) = 2\boldsymbol{a}_2,$$

it follows that

$$\boldsymbol{a}_2 = \frac{\boldsymbol{F}(t_0)}{2m}.$$

Mass therefore directly scales the first Taylor coefficient determined by the dynamical law.

The logical sequence is therefore

$$\text{positie} \longrightarrow \text{snelheid} \longrightarrow \text{massa als schaal} \longrightarrow \text{impuls} \longrightarrow \text{interactie als verandering van impuls.}$$

In this construction, mass is not a pre-given substance, but a property of matter that determines how much motion corresponds to a given velocity and how much an interaction changes that motion.

The momentum $\boldsymbol{p} = m\boldsymbol{v}$ specifies both the magnitude and the direction of the motion. A scalar quantity can be derived from this vector by contracting $\boldsymbol{p}$ with itself. We define the kinetic energy as

$$T = \frac{\boldsymbol{p} \cdot \boldsymbol{p}}{2m}.$$

Since $\boldsymbol{p} = m\boldsymbol{v}$, it follows that

$$T = \frac{m^2\boldsymbol{v} \cdot \boldsymbol{v}}{2m} = \frac{1}{2}m\|\boldsymbol{v}\|^2.$$

The direction of $\boldsymbol{v}$ no longer plays a role in this expression: only the magnitude $\|\boldsymbol{v}\|$ determines T. Whereas momentum describes the complete vectorial state of motion, T

describes only its scale. The transition from $\boldsymbol{p}$ to T is therefore a projection in which directional information is lost.

This loss of information is also visible in the dynamics. Differentiating T with respect to time gives

$$\frac{dT}{dt} = \frac{d}{dt}\left(\frac{1}{2}m\boldsymbol{v}\cdot\boldsymbol{v}\right) = m\,\boldsymbol{v}\cdot\frac{d\boldsymbol{v}}{dt} = m\boldsymbol{a}\cdot\boldsymbol{v}.$$

Using $\boldsymbol{F} = m\boldsymbol{a}$, it follows that

$$\frac{dT}{dt} = \boldsymbol{F}\cdot\boldsymbol{v}.$$

Only the component of $\boldsymbol{F}$ parallel to $\boldsymbol{v}$ therefore contributes to the change in T. If we decompose the force into a component parallel to the velocity and a component perpendicular to it,

$$\boldsymbol{F} = \boldsymbol{F}_{\parallel} + \boldsymbol{F}_{\perp}, \qquad \boldsymbol{F}_{\perp}\cdot\boldsymbol{v} = 0,$$

then

$$\frac{dT}{dt} = \boldsymbol{F}_{\parallel}\cdot\boldsymbol{v}.$$

The component $\boldsymbol{F}_{\perp}$ does change the direction of $\boldsymbol{p}$, and thus the trajectory, but leaves T unchanged. This is the case, for example, for a centripetal force: it keeps $\|\boldsymbol{v}\|$, and therefore T, constant while the direction of motion changes continuously.

Integrating $dT/dt = \boldsymbol{F}\cdot\boldsymbol{v}$ over time yields the work–energy theorem. For a trajectory between times t_1 and t_2,

$$T(t_2) - T(t_1) = \int_{t_1}^{t_2} \boldsymbol{F}\cdot\boldsymbol{v}\,dt = \int_{\boldsymbol{r}(t_1)}^{\boldsymbol{r}(t_2)} \boldsymbol{F}\cdot d\boldsymbol{r}.$$

Kinetic energy is therefore not an independently defined quantity, but the scalar scale derived from momentum by projecting out its direction. Whereas $\boldsymbol{p}$ records the complete change in motion, T records only the part of an interaction that acts parallel to the motion.

Just as kinetic energy was derived from $\boldsymbol{p}$, a second scalar quantity can be derived from momentum—now not from $\boldsymbol{p}$ itself, but from its change. With $\boldsymbol{F} = d\boldsymbol{p}/dt$, the work along a trajectory from $\boldsymbol{r}_0$ to $\boldsymbol{r}$ is

$$W = \int_{\boldsymbol{r}_0}^{\boldsymbol{r}} \boldsymbol{F} \cdot d\boldsymbol{r}' = \int_{t_0}^{t} \dot{\boldsymbol{p}} \cdot \boldsymbol{v}\, dt'.$$

When this integral is independent of the path followed between $\boldsymbol{r}_0$ and $\boldsymbol{r}$, that is, when

$$\nabla \times \boldsymbol{F} = \boldsymbol{0},$$

a scalar function of position can be defined:

$$U(\boldsymbol{r}) = -\int_{\boldsymbol{r}_0}^{\boldsymbol{r}} \boldsymbol{F} \cdot d\boldsymbol{r}'.$$

We call U the potential energy. This definition directly implies

$$\boldsymbol{F} = -\nabla U.$$

Potential energy is therefore not an independently introduced quantity, but a reformulation of the same $\boldsymbol{F} = \dot{\boldsymbol{p}}$ from which T was also derived. Whereas $T = \boldsymbol{p} \cdot \boldsymbol{p}/(2m)$ contracts the momentum at a single instant into a scale, U integrates the change in momentum, $\dot{\boldsymbol{p}}$, along the path into a scale:

$$T = \frac{\boldsymbol{p} \cdot \boldsymbol{p}}{2m} \qquad \text{versus} \qquad U = -\int \dot{\boldsymbol{p}} \cdot d\boldsymbol{r}.$$

Both quantities reduce the complete vectorial information in the trajectory $\boldsymbol{p}(t)$ to a single number, each in a different way: T by contracting $\boldsymbol{p}$ with itself at one instant, and U by integrating the change in $\boldsymbol{p}$ along the trajectory.

Because U is constructed from $\boldsymbol{F} = \dot{\boldsymbol{p}}$, and $\boldsymbol{F}$ describes the change in momentum of the object under consideration within its interaction with the environment, U is by construction a quantity of that interaction, not of the object in isolation. Equilibrium is a special case within this framework: $\boldsymbol{F} = \boldsymbol{0}$, and hence $\dot{\boldsymbol{p}} = \boldsymbol{0}$, corresponds to a critical point of U,

$$\nabla U = \boldsymbol{0},$$

but U itself is defined at every point of the trajectory, not only at equilibrium points.

From $dT/dt = \boldsymbol{F} \cdot \boldsymbol{v}$ and $\boldsymbol{F} = -\nabla U$, it follows that

$$\frac{dT}{dt} = -\nabla U \cdot \boldsymbol{v} = -\frac{dU}{dt}.$$

It follows directly that the sum of both quantities is constant:

$$\frac{d}{dt}(T + U) = 0.$$

We call $E = T + U$ the mechanical energy. Both terms are derived separately from the same underlying quantity $\boldsymbol{p}(t)$: T as a contraction of $\boldsymbol{p}$, and U as an integral of $\dot{\boldsymbol{p}}$. Conservation of their sum is therefore not an additional physical principle, but a direct consequence of the way in which both are constructed from the same trajectory $\boldsymbol{p}(t)$.

4 Relativistic Measure of Motion

In Newtonian mechanics, the relation between momentum and velocity is linear:

$$\frac{\boldsymbol{p}}{m} = \boldsymbol{v}.$$

Special relativity, however, shows that ordinary coordinate velocity is not the complete relativistic measure of motion. The relativistic momentum is

$$\boldsymbol{p} = \gamma(v) m \boldsymbol{v},$$

where

$$\gamma(v) = \frac{1}{\sqrt{1 - v^2/c^2}}$$

is the Lorentz factor.

The two factors m and $\gamma(v)$ have different meanings. The mass m is a constant intrinsic property of the material system. It determines the scale of the quantity of motion. The Lorentz factor $\gamma(v)$, by contrast, is not a property of matter itself, but a velocity-dependent factor arising from the geometry of spacetime.

The relativistic measure of motion can be written as

$$\boldsymbol{w} = \gamma(v)\boldsymbol{v}.$$

This quantity is called proper velocity or celerity. The relativistic momentum then once again takes the simple form

$$\boldsymbol{p} = m\boldsymbol{w}.$$

Thus, in special relativity as well, mass retains its meaning as a constant scale factor of motion. Only the appropriate measure of motion changes: in Newtonian mechanics it is $\boldsymbol{v}$, whereas in special relativity it is $\gamma\boldsymbol{v}$.

The construction can be summarized as follows:

$$\text{baan} \longrightarrow \text{kinematische snelheid} \longrightarrow \text{fysische bewegingsmaat} \longrightarrow \text{impuls}.$$

In Newtonian mechanics,

$$\text{bewegingsmaat} = \boldsymbol{v}, \qquad \boldsymbol{p} = m\boldsymbol{v}.$$

In special relativity,

$$\text{bewegingsmaat} = \gamma\boldsymbol{v}, \qquad \boldsymbol{p} = m\gamma\boldsymbol{v}.$$

In both theories, the mass m remains the constant intrinsic scale by which the physical measure of motion of matter is converted into momentum. The Lorentz factor γ does not belong to matter itself, but to the relativistic structure of motion in spacetime.

This description, however, remains limited to spatial momentum; energy does not yet appear in it. As noted in Section 1, time and space can be considered jointly in the worldline. To remain dimensionally consistent—all components must have the same unit—we now write it as

$$\boldsymbol{R}(\tau) = \big(ct(\tau), x(\tau), y(\tau), z(\tau)\big),$$

parametrized by proper time τ rather than coordinate time t. This is the same construction as $\boldsymbol{R}(t) = (t, x, y, z)$ in Section 1, with a factor c in the time component so that it is expressed in the same unit of length as x, y, z.

In Section 3, the three-dimensional momentum $\boldsymbol{p}$ was the primitive object from which T and U were derived as scalar reductions. When the worldline $\boldsymbol{R}(\tau)$ is taken as the primitive object, this order changes fundamentally. The four-velocity is defined as

$$u^\mu = \frac{dR^\mu}{d\tau} = \big(\gamma c, \gamma \boldsymbol{v}\big),$$

where $dt/d\tau = \gamma$ has been used. The spatial component of u^μ is precisely the proper velocity $\boldsymbol{w} = \gamma \boldsymbol{v}$ introduced above; the time component γc had no counterpart in the three-dimensional description.

The four-momentum is defined as

$$p^\mu = m u^\mu = \big(\gamma m c, \gamma m \boldsymbol{v}\big) = \left(\frac{E}{c}, \boldsymbol{p}\right),$$

where the time component is identified as E/c, with

$$E = \gamma m c^2.$$

In this four-dimensional framework, E and $\boldsymbol{p}$ are no longer a primitive quantity and a scalar reduction derived from it, as $\boldsymbol{p}$ and T were in Section 3. They are both components of the same object p^μ, just as x, y, z and ct are components of $\boldsymbol{R}$. No contraction or integration is required to pass from $\boldsymbol{p}$ to E; they are two projections of p^μ onto different axes.

The relation between these components follows from the invariant norm of p^μ:

$$p^\mu p_\mu = \left(\frac{E}{c}\right)^2 - \boldsymbol{p} \cdot \boldsymbol{p} = m^2 c^2,$$

or equivalently

$$E^2 = (pc)^2 + (mc^2)^2.$$

This relation is invariant under Lorentz transformations, in the same way that $\|\boldsymbol{R}\|$ was invariant under the Galilean transformation in Section 2—with the difference that space and time now occur jointly in the invariant.

The connection with Section 3 follows from the low-velocity limit. For $v \ll c$, the expansion is

$$\gamma = \left(1 - \frac{v^2}{c^2}\right)^{-1/2} \approx 1 + \frac{1}{2}\frac{v^2}{c^2},$$

so that

$$E = \gamma mc^2 \approx mc^2 + \frac{1}{2}mv^2 = mc^2 + T.$$

Classical kinetic energy T is therefore no longer an independently defined quantity, but the velocity-dependent part of E after subtracting the rest energy mc^2:

$$T \equiv E - mc^2.$$

In the worldline description, this T is therefore itself a projection—not of $\boldsymbol{p}$, as in Section 3, but of E, the time component of p^μ. Whether T is secondary to $\boldsymbol{p}$ or to E thus depends on whether space and time are considered separately (Section 3) or jointly as a worldline (this section). Both descriptions are correct within their own frameworks; the distinction between primitive and derived is itself dependent on which structure—$\mathbb{R}^3$ with external time, or the four-dimensional worldline—is taken as the point of departure.